THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10

Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 12	Take the lowest powers of common primes
2	**(a)** 29	$(\alpha+\beta)^2-2\alpha\beta=7^2-2(10)$
3	**(b)** -2	Eliminate y
4	**(a)** $\frac{9}{2}$	Sum $=-b/a$
5	**(b)** 24	Take the highest powers of all primes
6	**(b)** $\frac{5}{6}$	$\dfrac{\alpha+\beta}{\alpha\beta}$
7	**(d)** 5	Eliminate x
8	**(a)** 2	Product $=c/a$
9	**(b)** 13	HCF of 65 and 117
10	**(c)** 10	$2^2-7(2)+k=0$
11	**(b)** 2	All ratios equal
12	**(b)** 7	Roots are 2, 9
13	**(a)** 27	LCM + remainder
14	**(b)** 3	$2^2+2k-10=0$
15	**(a)** 12	$\frac{k}{a_2}\neq\frac{b_1}{b_2}$ for a unique solution

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Numbers exactly divisible: 240, 1024 (1); HCF $=16$ (1).	1+1
17	$\alpha+\beta=8$, $\alpha\beta=12$ (1); $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=40$ (1).	1+1
18	Write as $7x+2y+6=0$, $3x+5y-14=0$ (1); $\dfrac{x}{-58}=\dfrac{y}{116}=\dfrac{1}{29}$ (1); $x=-2$, $y=4$ (1).	1+1+1
19	$D=(-9)^2-4(2)(4)=49$ (1); $x=\dfrac{9\pm7}{4}$ (1); roots $\frac{1}{2}, 4$ (1).	1+1+1
20	LCM$(12,16,18)=144$ (1); number $=144+5=149$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	HCF × LCM = product of numbers (1); other $=\dfrac{9\times360}{72}$ (1) $=45$ (1).	1+1+1
22	$\alpha+\beta=6$, $\alpha\beta=8$ (1); $\dfrac{\alpha+\beta}{\alpha\beta}=\frac{3}{4}$ (1); $\alpha^2+\beta^2=20$ (1).	1+1+1
23	Multiply (i) by 5 and (ii) by 2 and subtract: $(29)x=-58$ (2); $x=-2$ (1); put in (i): $y=4$ (1); check in (ii) (1).	2+1+1+1
24	Multiply by $x(x+1)$: $2x+1=\frac{5}{6}x(x+1)$ (1); form the quadratic with integer coefficients (1); positive root $x=2$ (1).	1+1+1
	OR: Required time $=$ LCM$(12,15,18)$ (1); $12=2^2\times3$, $15=3\times5$, $18=2\times3^2$ (1); LCM $=180$ minutes (1).	1+1+1
25	Sum of new zeroes $=\dfrac{\alpha+\beta}{\alpha\beta}=\frac{5}{6}$ (1); product $=\dfrac{1}{\alpha\beta}=\frac{1}{6}$ (1); polynomial $x^2-(\frac{5}{6})x+(\frac{1}{6})$ (1).	1+1+1
	OR: Let tens digit x, ones digit y: $x+y=11$ (1); $(10y+x)-(10x+y)=45$ so $y-x=5$ (1); $x=3$, $y=8$, number 38 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Capacity $=$ HCF$(850,680)=170$ L (3); measures 5 and 4 (2).	3+2
	OR: Quadratic factor $(x-1)(x-2)=x^2-3x+2$ (1); divide to get $x-3$ (2); third zero 3 (1); zeroes $1, 2, 3$ (1).	1+2+1+1

27	Son x, father $10x$ (1); $10x + 6 = 4(x + 6)$ (2); $x = 3$ (1); son 3 years, father 30 years (1).	1+2+1+1
	OR: Speed x: $\frac{200}{x} - \frac{200}{x+5} = 2$ (1); $1000 = 2x(x + 5)$ so $x^2 + 5x - 500 = 0$ (2); $(x - 20)(x + 25) = 0$ (1); speed 20 km/h (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 20	$16 - 36 + k = 0$
2	**(b)** non-terminating repeating	$210 = 2 \times 3 \times 5 \times 7$
3	**(b)** 14	Remainder $= p(2) = 14$
4	**(d)** 9	Add the two equations
5	**(c)** two distinct real roots	$D = 4$
6	**(c)** rational	$\sqrt{3} \times \sqrt{3} = 3$
7	**(b)** 4	Other zero $= 7 - 3$
8	**(b)** 27	$(x+y)(x-y) = 9 \times 3$
9	**(c)** 8	Factorise
10	**(a)** 15	Take the lowest powers of common primes
11	**(a)** 25	$(\alpha+\beta)^2 - 2\alpha\beta = 7^2 - 2(12)$
12	**(d)** 4	Eliminate y
13	**(d)** $-\frac{7}{3}$	Sum $= -b/a$
14	**(c)** 60	Take the highest powers of all primes
15	**(d)** $\frac{5}{4}$	$\frac{\alpha+\beta}{\alpha\beta}$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$p(3) = 0$: $-15 + k = 0$ (1) so $k = 15$ (1).	1+1
17	Unique solution when $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ (1): $\frac{k}{6} \neq \frac{2}{1}$, i.e. $k \neq 12$ (1).	1+1
18	Put $x = 4$: $16 - 36 + k = 0$ (1); $k = 20$ (1); other root $= 9 - 4 = 5$ (1).	1+1+1
19	$200 = 2^3 \times 5^2$ (1); places $= \max$ of the powers of 2 and 5 $= 3$ (1).	1+1
20	New sum $= 2(3) = 6$, product $= 4(2) = 8$ (1); polynomial $x^2 - 6x + 8$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$D = 25$ (1); two distinct real roots (1); roots $2, \frac{1}{3}$ (1).	1+1+1
22	Product $= 5^2 - 4 \times 3 = 13$ is rational (2); $5 + 2\sqrt{3}$ is irrational as sum of rational and irrational (1).	2+1
23	$p(6) = 0$: $36 - 66 + 30 = 0$ gives $k = -11$ (1); sum of zeroes $= -k = 11$ (1); other zero $= 5$ (1).	1+1+1
24	$\frac{k}{4} = \frac{1}{k}$ gives $k^2 = 4$ (1); $k = \pm 2$ (1); only $k = 2$ also satisfies $\frac{1}{k} = \frac{k-1}{k}$ (1).	1+1+1
	OR: Divide by 1: $x^2 - 10x + 21 = 0$ (1); $(x-5)^2 = 4$ (1); $x - 5 = \pm 2$, so $x = 3, 7$ (1).	1+1+1
25	HCF × LCM = product of numbers (1); other $= \frac{12 \times 504}{84}$ (1) $= 72$ (1).	1+1+1
	OR: $\alpha + \beta = 6$, $\alpha\beta = 5$ (1); $\frac{\alpha+\beta}{\alpha\beta} = \frac{6}{5}$ (1); $\alpha^2 + \beta^2 = 26$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Side $= \text{HCF}(24, 16) = 8$ m (2); tiles $= \frac{24 \times 16}{64} = 6$ (3).	2+3
	OR: Sum $= 5$, product $= k$ (1); $k = 2 \times 5 = 10$ (2); polynomial $x^2 - 5x + 10$ (2).	1+2+2
27	Let fraction $\frac{x}{y}$: $2(x+1) = 1(y+1)$ and $9(x+2) = 5(y+2)$ (2); simplify and solve (2); $x = 3$, $y = 7$, fraction $\frac{3}{7}$ (1).	2+2+1

	OR: Base x, altitude $x-1$: $x^2+(x-1)^2=841$ (1); $2x^2-2x+-840=0$, i.e. $x^2-1x-420=0$ (2); $(x-21)(x+20)=0$ (1); base 21 cm, altitude 20 cm (1).	1+2+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 3	Eliminate x
2	**(a)** -2	Product $= c/a$
3	**(c)** 1	HCF of 55 and 96
4	**(a)** 15	$3^2 - 8(3) + k = 0$
5	**(c)** 3	All ratios equal
6	**(d)** 8	Roots are -3, 5
7	**(c)** 62	LCM + remainder
8	**(c)** -1	$3^2 + 3k - 6 = 0$
9	**(a)** 12	$\frac{k}{a_2} \neq \frac{b_1}{b_2}$ for a unique solution
10	**(d)** 30	$25 - 55 + k = 0$
11	**(a)** terminating	$312500 = 2^2 \times 5^7$
12	**(b)** 6	Remainder $= p(1) = 6$
13	**(c)** 8	Add the two equations
14	**(c)** two equal real roots	$D = 0$
15	**(d)** rational	$\sqrt{5} \times \sqrt{5} = 5$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$\frac{2}{4} = \frac{k}{4} = \frac{6}{12}$ (1) gives $k = 2$ (1).	1+1
17	$D = 0$: $(6)^2 - 4k(1) = 0$ (1); $k = 9$ (1).	1+1
18	Numbers exactly divisible: 391, 425 (1); HCF $= 17$ (1).	1+1
19	$\alpha + \beta = 10$, $\alpha\beta = 21$ (1); $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = 58$ (1).	1+1
20	Write as $5x + 3y - 29 = 0$, $4x + 7y - 37 = 0$ (1); $\frac{x}{92} = \frac{y}{69} = \frac{1}{23}$ (1); $x = 4$, $y = 3$ (1).	1+1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Multiply (i) by 7 and (ii) by 3 and subtract: $(23)x = 92$ (2); $x = 4$ (1); put in (i): $y = 3$ (1); check in (ii) (1).	2+1+1+1
22	Multiply by $x(x+1)$: $2x + 1 = \frac{7}{12}x(x+1)$ (1); form the quadratic with integer coefficients (1); positive root $x = 3$ (1).	1+1+1
23	Required time $=$ LCM$(6, 10, 14)$ (1); $6 = 2 \times 3$, $10 = 2 \times 5$, $14 = 2 \times 7$ (1); LCM $= 210$ minutes (1).	1+1+1
24	Sum of new zeroes $= \frac{\alpha+\beta}{\alpha\beta} = \frac{7}{6}$ (1); product $= \frac{1}{\alpha\beta} = \frac{1}{6}$ (1); polynomial $x^2 - (\frac{7}{6})x + (\frac{1}{6})$ (1).	1+1+1
	OR: Let tens digit x, ones digit y: $x + y = 13$ (1); $(10y + x) - (10x + y) = 45$ so $y - x = 5$ (1); $x = 4$, $y = 9$, number 49 (1).	1+1+1
25	$D = -7$ (1); no real roots (1); no real roots exist (1).	1+1+1
	OR: Product $= 7^2 - 9 \times 2 = 31$ is rational (2); $7 + 3\sqrt{2}$ is irrational as sum of rational and irrational (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Capacity $=$ HCF$(612, 918) = 306$ L (3); measures 2 and 3 (2).	3+2
	OR: Quadratic factor $(x+1)(x-2) = x^2 - x - 2$ (1); divide to get $x - 5$ (2); third zero 5 (1); zeroes $-1, 2, 5$ (1).	1+2+1+1
27	Son x, father $6x$ (1); $6x + 10 = 5(x + 10)$ (2); $x = 40$ (1); son 40 years, father 240 years (1).	1+2+1+1

	OR: Speed x: $\frac{280}{x} - \frac{280}{x+5} = 1$ (1); $1400 = 1x(x+5)$ so $x^2 + 5x - 1400 = 0$ (2); $(x-35)(x+40) = 0$ (1); speed 35 km/h (1).	1+2+1+1

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